IT13: ADBMS

Chapter 2:SQL

Practical Assignment 1 Solution Template

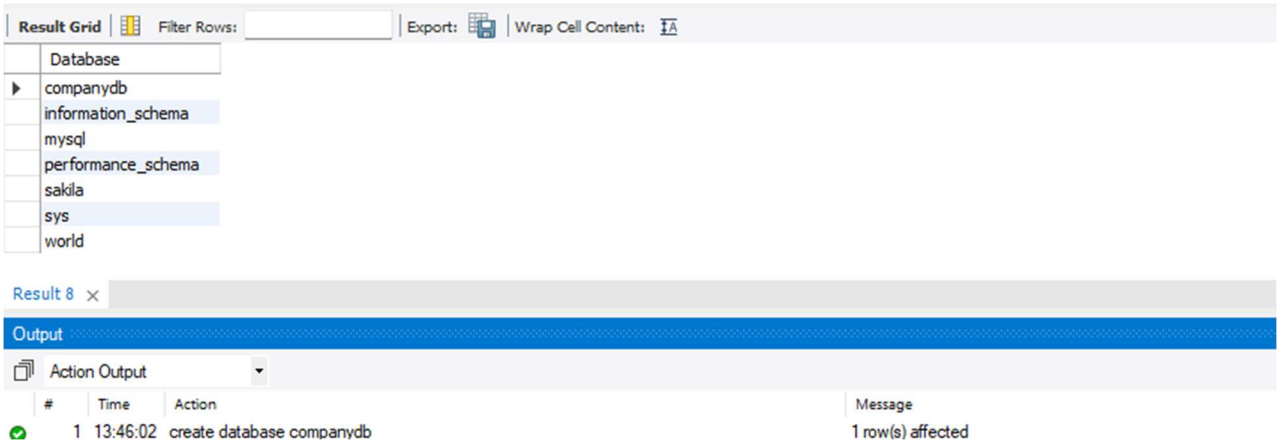
Name : Sanket Jagdish Patil

Roll No : 2601144

Write the SQL query and paste a screenshot of the output below the question.

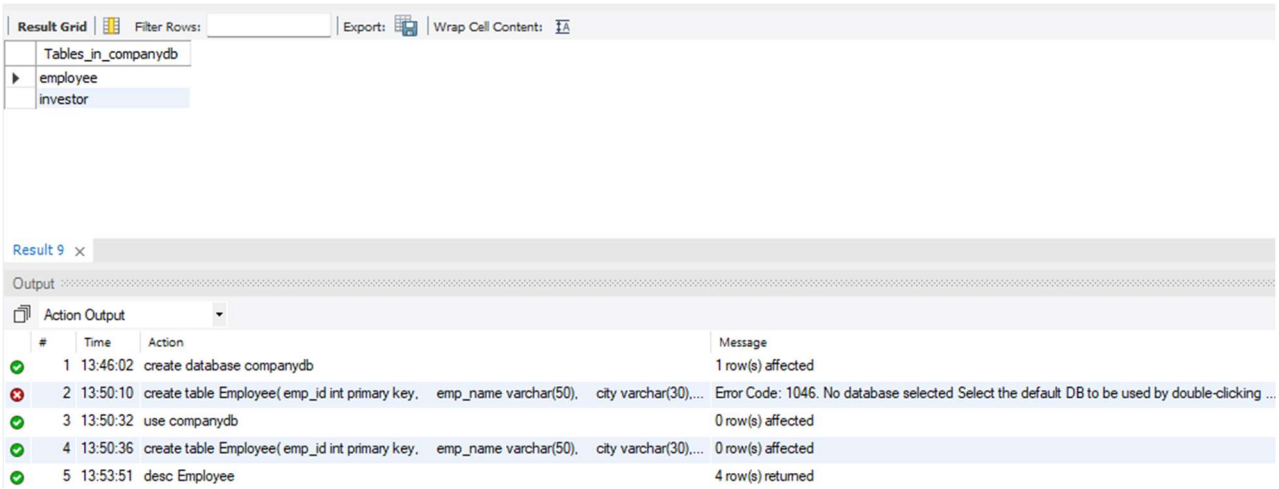
1. Create a database and two tables in it.

```
create database companydb;
use companydb;
show databases;
create table Employee(
        emp_id int primary key,
   emp_name varchar(50),
   city varchar(30),
   bdate date
   );
create table Investor(
        inv_no int primary key,
   inv_name varchar (50),
   inv_date date,
   inv_amt int,
   emp_id int
   );
```



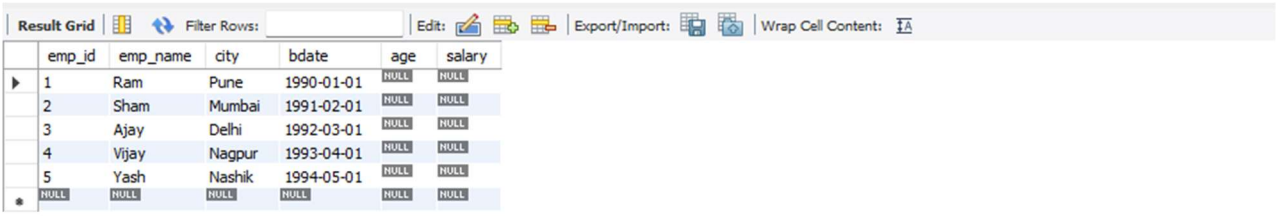
2. Insert at least 5 rows into the table.

```
insert into Employee values(1,'Ram','Pune','1990-01-01');
insert into Employee values(2,'Sham','Mumbai','1991-02-01');
insert into Employee values(3,'Ajay','Delhi','1992-03-01');
insert into Employee values(4,'Vijay','Nagpur','1993-04-01');
insert into Employee values(5,'Yash','Nashik','1994-05-01');
insert into Investor values(11,'Bob','2020-01-01',10000,1);
insert into Investor values(12,'Jack','2020-02-01',20000,2);
insert into Investor values(13,'Logan','2020-03-01',30000,3);
insert into Investor values(14,'Tom','2020-04-01',40000,4);
insert into Investor values(15,'Patt','2020-05-01',50000,5);
```



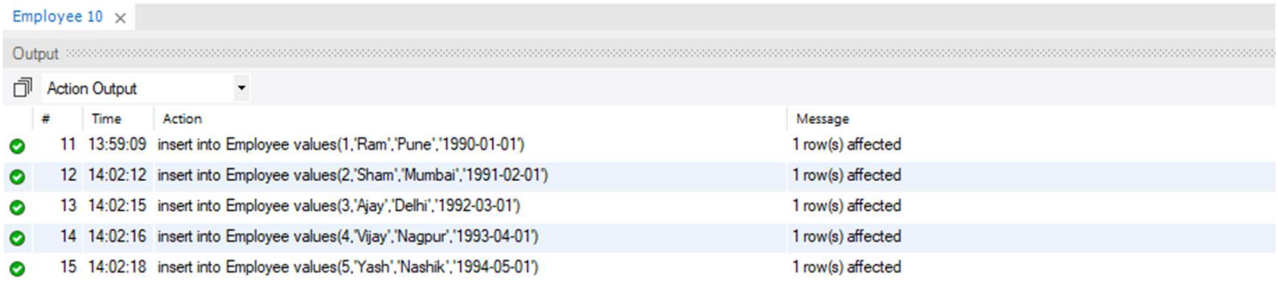
3. Display all the information in the Employee table.

desc employee;



Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:

emp_id	emp_name	city	bdate	age	salary
1	Ram	Pune	1990-01-01	NULL	NULL
2	Sham	Mumbai	1991-02-01	NULL	NULL
3	Ajay	Delhi	1992-03-01	NULL	NULL
4	Vijay	Nagpur	1993-04-01	NULL	NULL
5	Yash	Nashik	1994-05-01	NULL	NULL
NULL	NULL	NULL	NULL	NULL	NULL



Employee 10

Output

Action Output

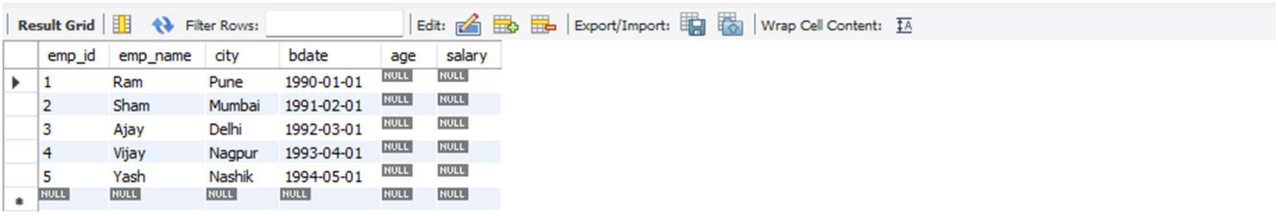
#	Time	Action	Message
11	13:59:09	insert into Employee values(1,'Ram','Pune','1990-01-01')	1 row(s) affected
12	14:02:12	insert into Employee values(2,'Sham','Mumbai','1991-02-01')	1 row(s) affected
13	14:02:15	insert into Employee values(3,'Ajay','Delhi','1992-03-01')	1 row(s) affected
14	14:02:16	insert into Employee values(4,'Vijay','Nagpur','1993-04-01')	1 row(s) affected
15	14:02:18	insert into Employee values(5,'Yash','Nashik','1994-05-01')	1 row(s) affected

4. Add columns 'Age' and 'Salary' to the Employee table; apply check constraints for (Age<=50 and Salary>=10000).

alter table Employee add column age int;

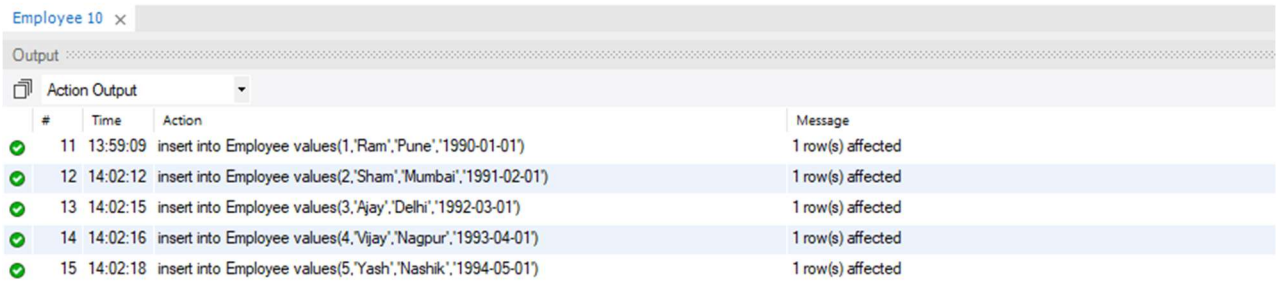
alter table Employee add column salary int;

alter table Employee add constraint check_age_salary check(age<=50 and salary>=10000);



Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:

emp_id	emp_name	city	bdate	age	salary
1	Ram	Pune	1990-01-01	NULL	NULL
2	Sham	Mumbai	1991-02-01	NULL	NULL
3	Ajay	Delhi	1992-03-01	NULL	NULL
4	Vijay	Nagpur	1993-04-01	NULL	NULL
5	Yash	Nashik	1994-05-01	NULL	NULL
NULL	NULL	NULL	NULL	NULL	NULL



Employee 10

Output

Action Output

#	Time	Action	Message
11	13:59:09	insert into Employee values(1,'Ram','Pune','1990-01-01')	1 row(s) affected
12	14:02:12	insert into Employee values(2,'Sham','Mumbai','1991-02-01')	1 row(s) affected
13	14:02:15	insert into Employee values(3,'Ajay','Delhi','1992-03-01')	1 row(s) affected
14	14:02:16	insert into Employee values(4,'Vijay','Nagpur','1993-04-01')	1 row(s) affected
15	14:02:18	insert into Employee values(5,'Yash','Nashik','1994-05-01')	1 row(s) affected

5. Add a column 'Age' to the Investor table.

alter table Investor add column Age int;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

inv_no	inv_name	inv_date	inv_amt	emp_id	Age
11	Bob	2020-01-01	10000	1	NULL
12	Jack	2020-02-01	20000	2	NULL
13	Logan	2020-03-01	30000	3	NULL
14	Tom	2020-04-01	40000	4	NULL
15	Patt	2020-05-01	50000	5	NULL
NULL	NULL	NULL	NULL	NULL	NULL

Investor 11 ×

Output

Action Output

#	Time	Action	Message
44	14:44:17	show databases	7 row(s) returned
45	14:45:50	show tables	2 row(s) returned
46	14:47:20	select * from Employee LIMIT 0, 1000	5 row(s) returned
47	14:51:01	select * from investors LIMIT 0, 1000	Error Code: 1146. Table 'companydb.investors' doesn't exist
48	14:51:11	select * from investors LIMIT 0, 1000	Error Code: 1146. Table 'companydb.investors' doesn't exist
49	14:51:49	select * from Investor LIMIT 0, 1000	5 row(s) returned

6. Change the data type of inv_name from varchar to text in the Investor table

   alter table Investor modify column inv_name text;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
inv_no	int	NO	PRI	NULL	
inv_name	text	YES		NULL	
inv_date	date	YES		NULL	
inv_amt	int	YES		NULL	
emp_id	int	YES		NULL	
Age	int	YES		NULL	

Result 12 ×

Output

Action Output

#	Time	Action	Message
46	14:47:20	select * from Employee LIMIT 0, 1000	5 row(s) returned
47	14:51:01	select * from investors LIMIT 0, 1000	Error Code: 1146. Table 'companydb.investors' doesn't exist
48	14:51:11	select * from investors LIMIT 0, 1000	Error Code: 1146. Table 'companydb.investors' doesn't exist
49	14:51:49	select * from Investor LIMIT 0, 1000	5 row(s) returned
50	14:59:15	alter table Investor modify column inv_name text	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
51	14:59:35	desc investor	6 row(s) returned

7. Change the name of the column/field Emp_id to E_id.

   alter table Employee rename column emp_id to e_id;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
e_id	int	NO	PRI	NULL	
emp_name	varchar(50)	YES		NULL	
city	varchar(30)	YES		NULL	
bdate	date	YES		NULL	
age	int	YES		NULL	
salary	int	YES		NULL	

Result 13

Output

Action Output

#	Time	Action	Message
48	14:51:11	select * from investors LIMIT 0, 1000	Error Code: 1146. Table 'companydb.investors' doesn't exist
49	14:51:49	select * from Investor LIMIT 0, 1000	5 row(s) returned
50	14:59:15	alter table Investor modify column inv_name text	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
51	14:59:35	desc investor	6 row(s) returned
52	15:04:46	alter table Employee rename column emp_id to e_id	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
53	15:04:59	desc employee	6 row(s) returned

8. Update the city of Emp_id-4 to the current city, Nagpur.

```
update employee set city='Nagpurr'
where e_id = 4;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

e_id	emp_name	city	bdate	age	salary
1	Ram	Pune	1990-01-01	NULL	NULL
2	Sham	Mumbai	1991-02-01	NULL	NULL
3	Ajay	Delhi	1992-03-01	NULL	NULL
4	Vijay	Nagpurr	1993-04-01	NULL	NULL
5	Yash	Nashik	1994-05-01	NULL	NULL
NULL	NULL	NULL	NULL	NULL	NULL

employee 14

Output

Action Output

#	Time	Action	Message
52	15:04:46	alter table Employee rename column emp_id to e_id	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
53	15:04:59	desc employee	6 row(s) returned
54	15:11:30	update employee set city='Nagpurr' where e_id = 4	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
55	15:12:15	select * from table employee	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds ..
56	15:12:19	select * from table employees	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds ..
57	15:12:25	select * from employee LIMIT 0, 1000	5 row(s) returned

9. Delete one record from the Investor table.

```
delete  from investor
where inv_no=14;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

inv_no	inv_name	inv_date	inv_amt	emp_id	Age
11	Bob	2020-01-01	10000	1	NULL
12	Jack	2020-02-01	20000	2	NULL
13	Logan	2020-03-01	30000	3	NULL
15	Patt	2020-05-01	50000	5	NULL
NULL	NULL	NULL	NULL	NULL	NULL

investor 16

Output

Action Output

#	Time	Action	Message
58	15:15:09	delete from investor where e_id=4	Error Code: 1054. Unknown column 'e_id' in 'where clause'
59	15:15:20	delete from investor where inv_id=4	Error Code: 1054. Unknown column 'inv_id' in 'where clause'
60	15:15:34	delete from investor where inv_no=4	0 row(s) affected
61	15:16:05	select * from investor LIMIT 0, 1000	5 row(s) returned
62	15:16:18	delete from investor where inv_no=14	1 row(s) affected
63	15:16:21	select * from investor LIMIT 0, 1000	4 row(s) returned

10. Drop the column 'Age' from the Investor table.

    alter table investor drop column age;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
inv_no	int	NO	PRI	NULL	
inv_name	text	YES		NULL	
inv_date	date	YES		NULL	
inv_amt	int	YES		NULL	
emp_id	int	YES		NULL	

Result 17

Output

Action Output

#	Time	Action	Message
60	15:15:34	delete from investor where inv_no=4	0 row(s) affected
61	15:16:05	select * from investor LIMIT 0, 1000	5 row(s) returned
62	15:16:18	delete from investor where inv_no=14	1 row(s) affected
63	15:16:21	select * from investor LIMIT 0, 1000	4 row(s) returned
64	15:18:46	alter table investor drop column age	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
65	15:19:01	desc investor	5 row(s) returned

11. Drop the CHECK constraint on the 'Age' column from the Employee table.

    alter table employee drop constraint check_age_salary;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
e_id	int	NO	PRI	NULL	
emp_name	varchar(50)	YES		NULL	
city	varchar(30)	YES		NULL	
bdate	date	YES		NULL	
age	int	YES		NULL	
salary	int	YES		NULL	

Result 19

Output

Action Output

#	Time	Action	Message
63	15:16:21	select * from investor LIMIT 0, 1000	4 row(s) returned
64	15:18:46	alter table investor drop column age	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
65	15:19:01	desc investor	5 row(s) returned
66	15:24:12	select inv_no,inv_name,inv_date,inv_amt from investor LIMIT 0, 1000	4 row(s) returned
67	15:26:35	alter table employee drop constraint check_age_salary	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
68	15:28:07	desc employee	6 row(s) returned

12. Create foreign key constraints between the Employee and Investor tables

    alter table investor add constraint foreign key(emp_id) references Employee(E_id);

    desc Investor;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
inv_no	int	NO	PRI	NULL	
inv_name	text	YES		NULL	
inv_date	date	YES		NULL	
inv_amt	int	YES		NULL	
emp_id	int	YES	MUL	NULL	

Result 20

Output

Action Output

#	Time	Action	Message
65	15:19:01	desc investor	5 row(s) returned
66	15:24:12	select inv_no,inv_name,inv_date,inv_amt from investor LIMIT 0, 1000	4 row(s) returned
67	15:26:35	alter table employee drop constraint check_age_salary	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
68	15:28:07	desc employee	6 row(s) returned
69	15:30:22	alter table investor add constraint foreign key(emp_id) references Employee(E_id)	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
70	15:30:47	desc investor	5 row(s) returned

13. Display any 4 columns from the Investor table.

    select inv_no,inv_name,inv_date,inv_amt from investor;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

inv_no	inv_name	inv_date	inv_amt
11	Bob	2020-01-01	10000
12	Jack	2020-02-01	20000
13	Logan	2020-03-01	30000
15	Patt	2020-05-01	50000
NULL	NULL	NULL	NULL

investor 18

Output

Action Output

#	Time	Action	Message
61	15:16:05	select * from investor LIMIT 0, 1000	5 row(s) returned
62	15:16:18	delete from investor where inv_no=14	1 row(s) affected
63	15:16:21	select * from investor LIMIT 0, 1000	4 row(s) returned
64	15:18:46	alter table investor drop column age	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
65	15:19:01	desc investor	5 row(s) returned
66	15:24:12	select inv_no,inv_name,inv_date,inv_amt from investor LIMIT 0, 1000	4 row(s) returned

14. Rename a table.

    rename table employee to newemployee;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Tables_in_companydb
investor
newemployee

Result 21

Output

Action Output

#	Time	Action	Message
67	15:26:35	alter table employee drop constraint check_age_salary	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
68	15:28:07	desc employee	6 row(s) returned
69	15:30:22	alter table investor add constraint foreign key(emp_id) references Employee(E_id)	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
70	15:30:47	desc investor	5 row(s) returned
71	15:33:39	rename table employee to newemployee	0 row(s) affected
72	15:34:54	show tables	2 row(s) returned

15. Delete all the records using the TRUNCATE command.

    truncate table newemployee;

    truncate table investor;

74	15:36:27	truncate table investor	0 row(s) affected

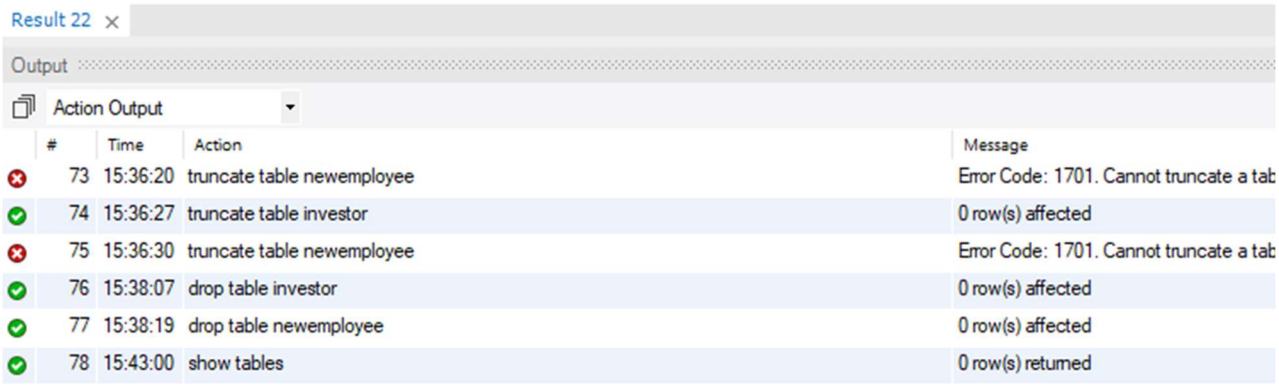
16. Drop both tables from the database.

```
drop table investor;
drop table newemployee;
```



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Tables_in_companydb



Result 22

Output

Action Output

#	Time	Action	Message
73	15:36:20	truncate table newemployee	Error Code: 1701. Cannot truncate a tab
74	15:36:27	truncate table investor	0 row(s) affected
75	15:36:30	truncate table newemployee	Error Code: 1701. Cannot truncate a tab
76	15:38:07	drop table investor	0 row(s) affected
77	15:38:19	drop table newemployee	0 row(s) affected
78	15:43:00	show tables	0 row(s) returned